INCH-POUND
MIL-DTL-16377/53E
14 December 2022
SUPERSEDING
MIL-DTL-16377/53D
30 June 2014

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; LIGHT EMITTING DIODE (LED), DETAIL LIGHTING, LANTERN, HAND, PORTABLE AND BULKHEAD-MOUNTED, WATERTIGHT, SYMBOLS 100.2L, 101.2L, AND 101.3L

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise specified, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Incandescent lanterns have been removed from this specification sheet in accordance with direction by the Navy Chief Engineer that all new and replacement light fixtures use light emitting diodes (LED) to achieve improved efficiency and shipboard personnel quality of life.

Rechargeable portable lanterns do not offer any advantage over legacy non-rechargeable portable lanterns and thus have been removed from this specification sheet.



PORTABLE

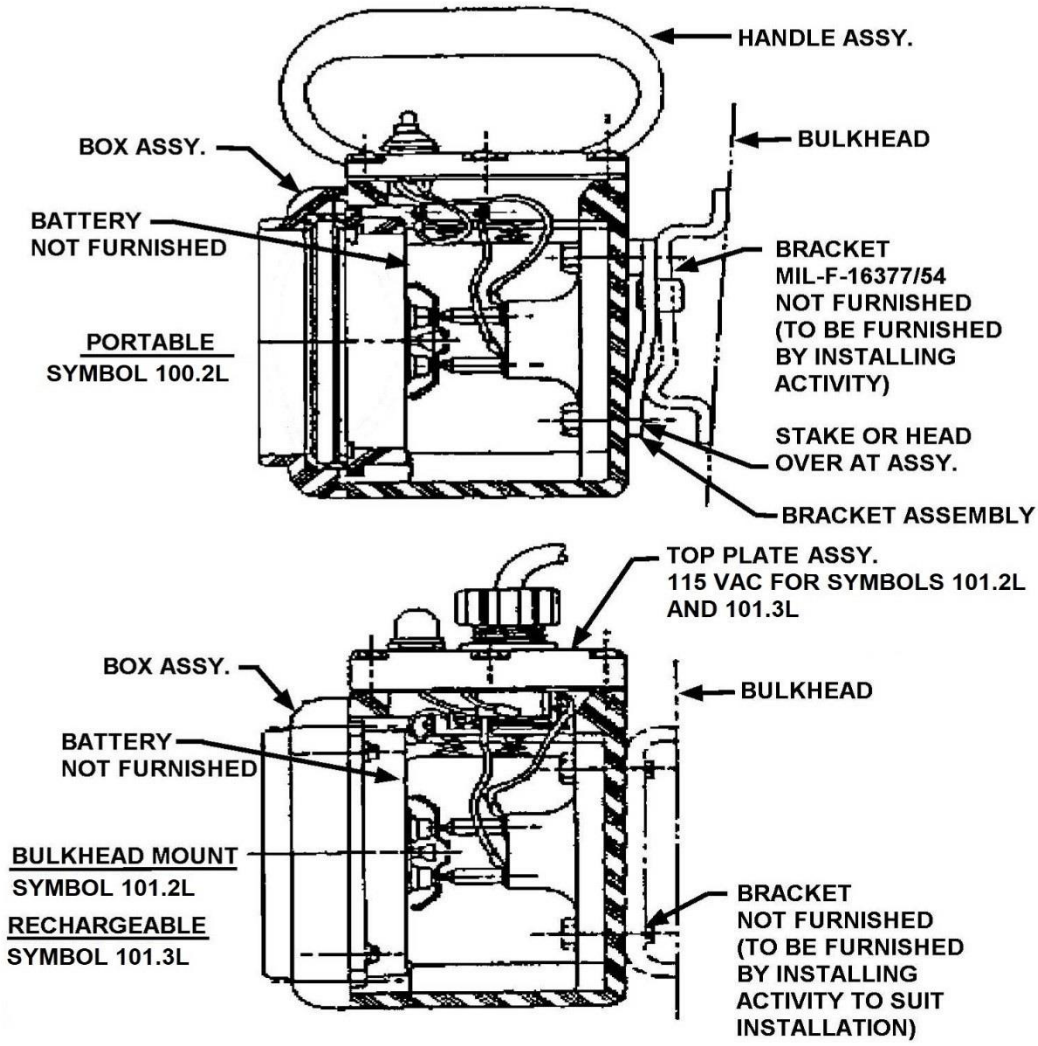


HARD WIRED/BULKHEAD

FIGURE 1. Lanterns, hand, portable and hard wired/bulkhead-mounted.

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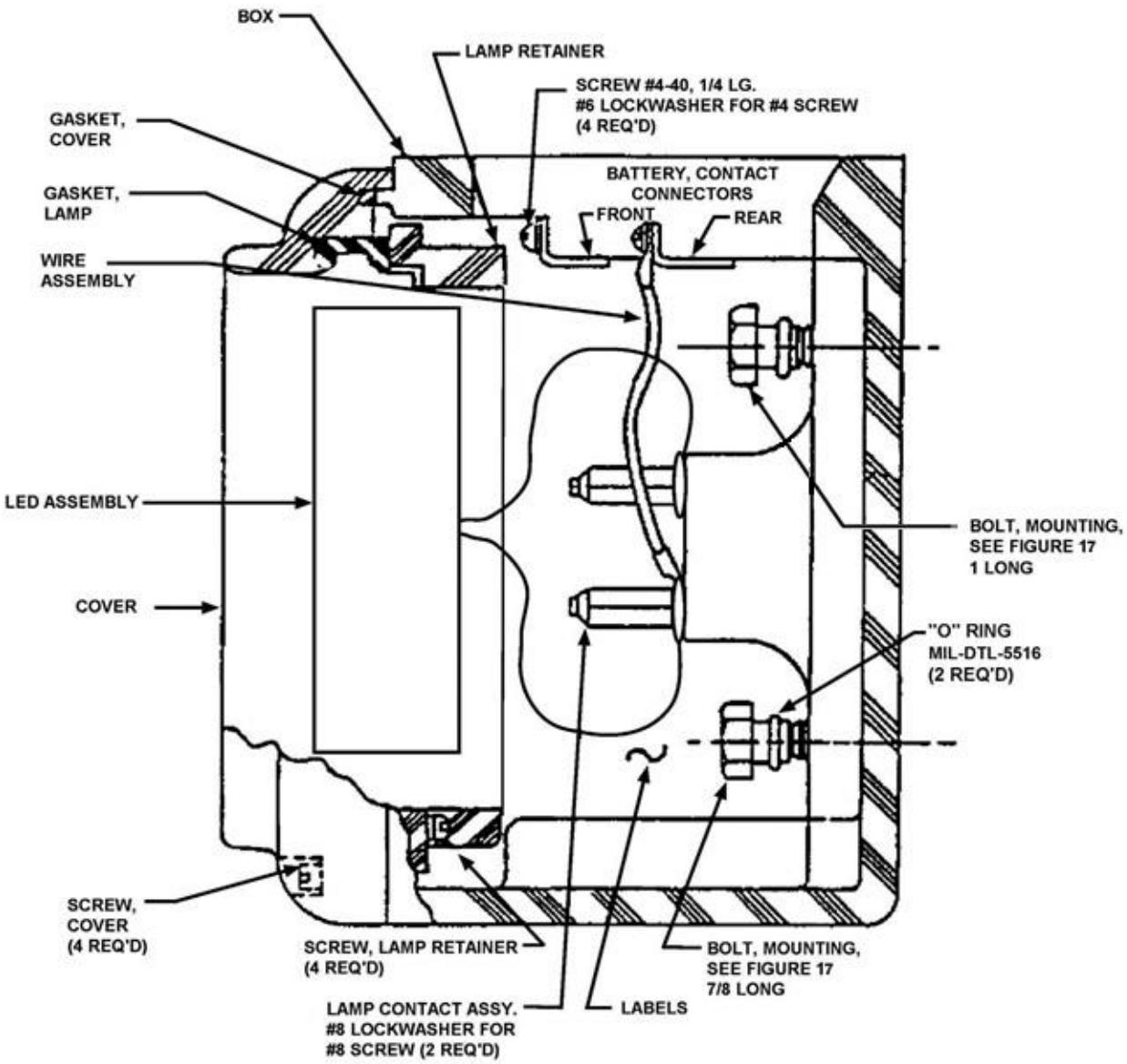
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NOTE: Symbols 100.2L, 101.2L, and 101.3L with LED assemblies.

FIGURE 2. Lantern assembly and typical mounting.

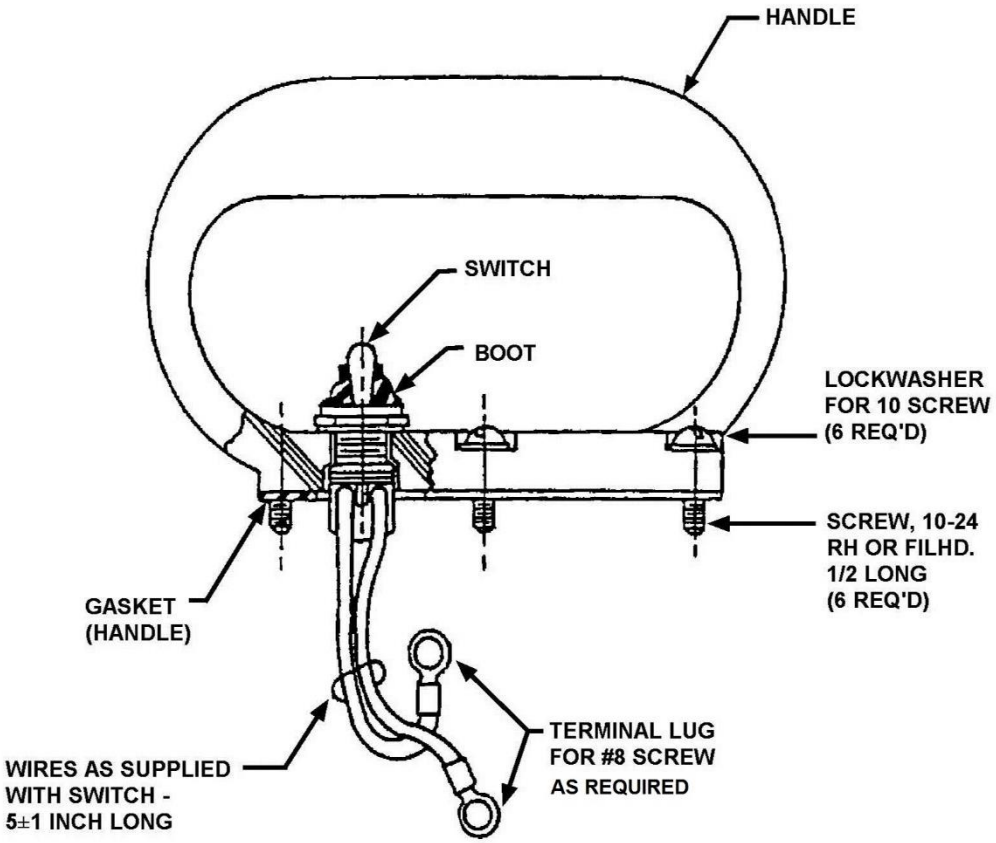
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NOTE: The connection of the LED assembly to the legacy incandescent lamp contacts (posts) in the figure is notional. Manufacturers may make this connection in whatever way they see fit, as long as the connection is made.

FIGURE 3. LED body assembly.

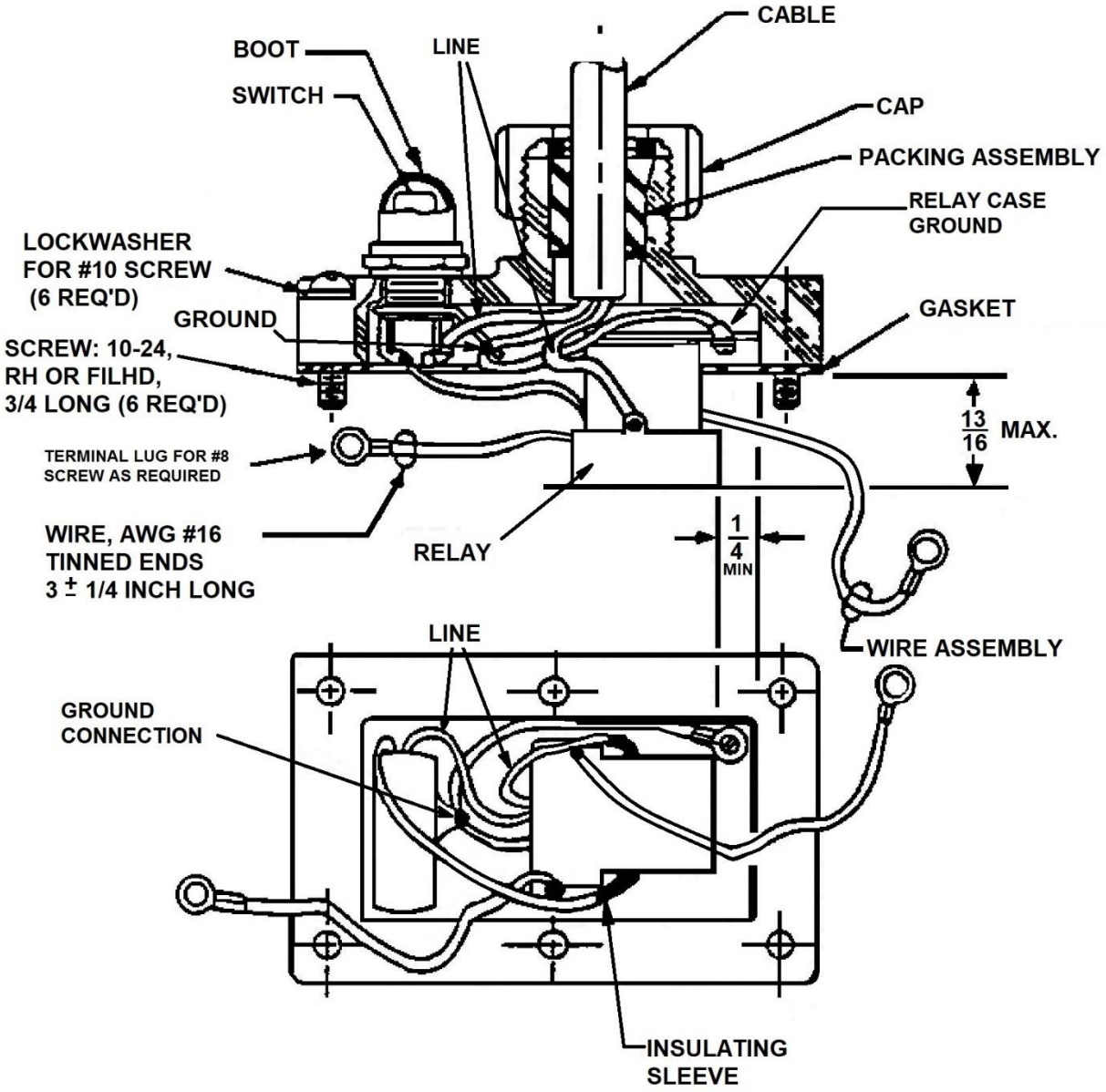
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NOTES:
1. See figure 19 for wiring diagram.
2. Wire lengths in this figure are nominal. Manufacturers shall ensure that wires are long enough to make internal connections.

FIGURE 4. Handle assembly for symbol 100.2L.

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NOTE: Relay ground required for metallic case relays only.

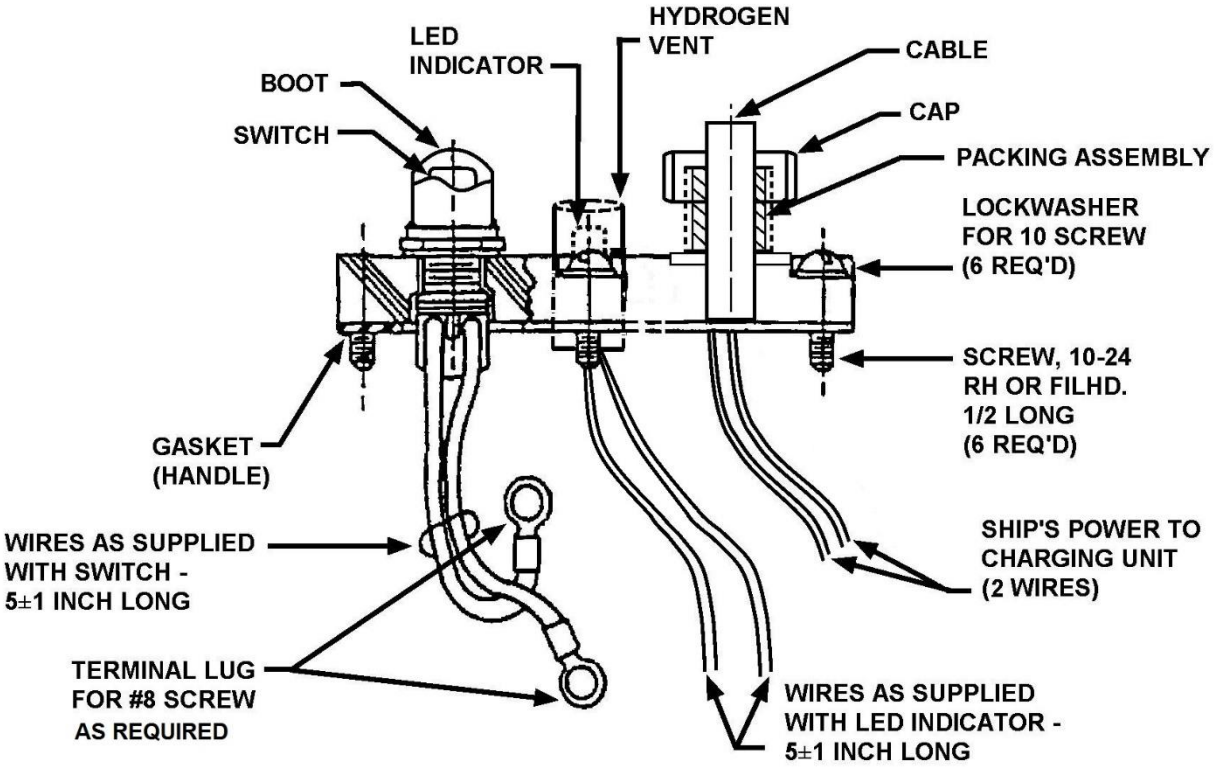
FIGURE 5. Bulkhead-mounted top plate assembly for symbol 101.2L.

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TABLE 1. Relay requirements, symbol 101.2L.

Characteristic	Requirement
Frequency	60 hertz
Duty	Continuous
Relay configuration	Single pole, double throw
Nominal coil voltage	110-125 VAC
Maximum allowable voltage	132 VAC minimum
Pull-in voltage	80-100 VAC
Drop out voltage	<40 VAC
Contact rating	10 amperes minimum, 240 VAC minimum
Service life	100,000 cycles minimum
Contact resistance	0.1 ohm maximum
Insulation resistance	100 megohms minimum
Dielectric strength	1,000 Vrms minimum
Dimensions	1.75 x 1.35 x 1.35 (maximum)

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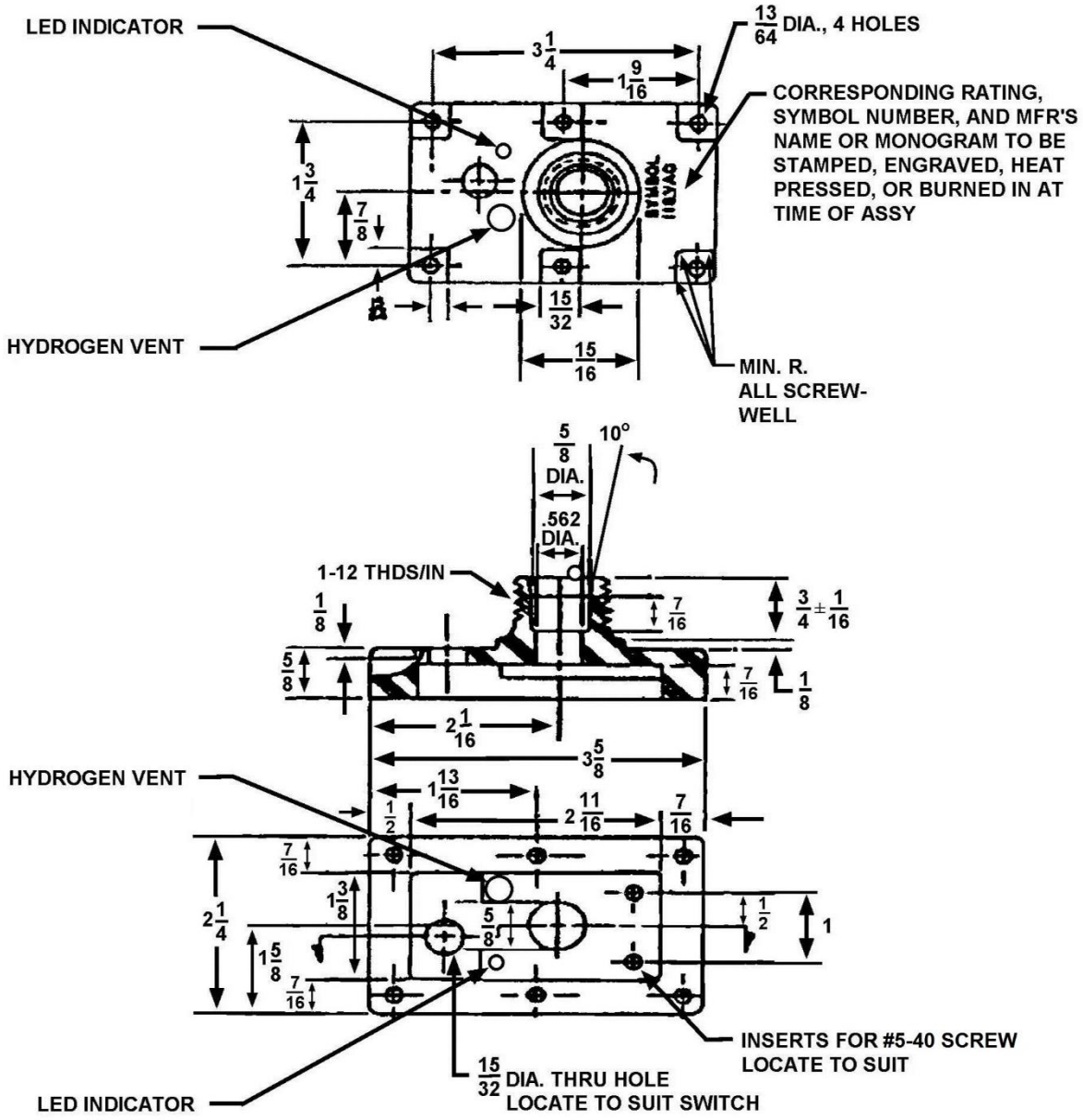


NOTES:

1. Test switch shall also be included. Switch shall be watertight.
2. Location of test switch, LED indicator, and hydrogen vent is at the discretion of the manufacturer.
3. Wire lengths in this figure are nominal. Manufacturers shall ensure that wires are long enough to make internal connections.
4. Incoming cable contains three conductors. The third conductor is used to provide a safety ground for the pushbutton switch and any other exposed metal parts. This conductor is not shown in the drawing to improve clarity.

FIGURE 6. Bulkhead-mounted top plate assembly for symbol 101.3L.

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NOTES:

1. Test switch shall also be included.
2. Location of test switch, LED indicator, and hydrogen vent is at the discretion of the manufacturer. LED indicator and hydrogen vent shall not be present for symbol 101.2L top plate assembly.

FIGURE 7. Bulkhead-mounted top plate assembly for symbols 101.2L and 101.3L.

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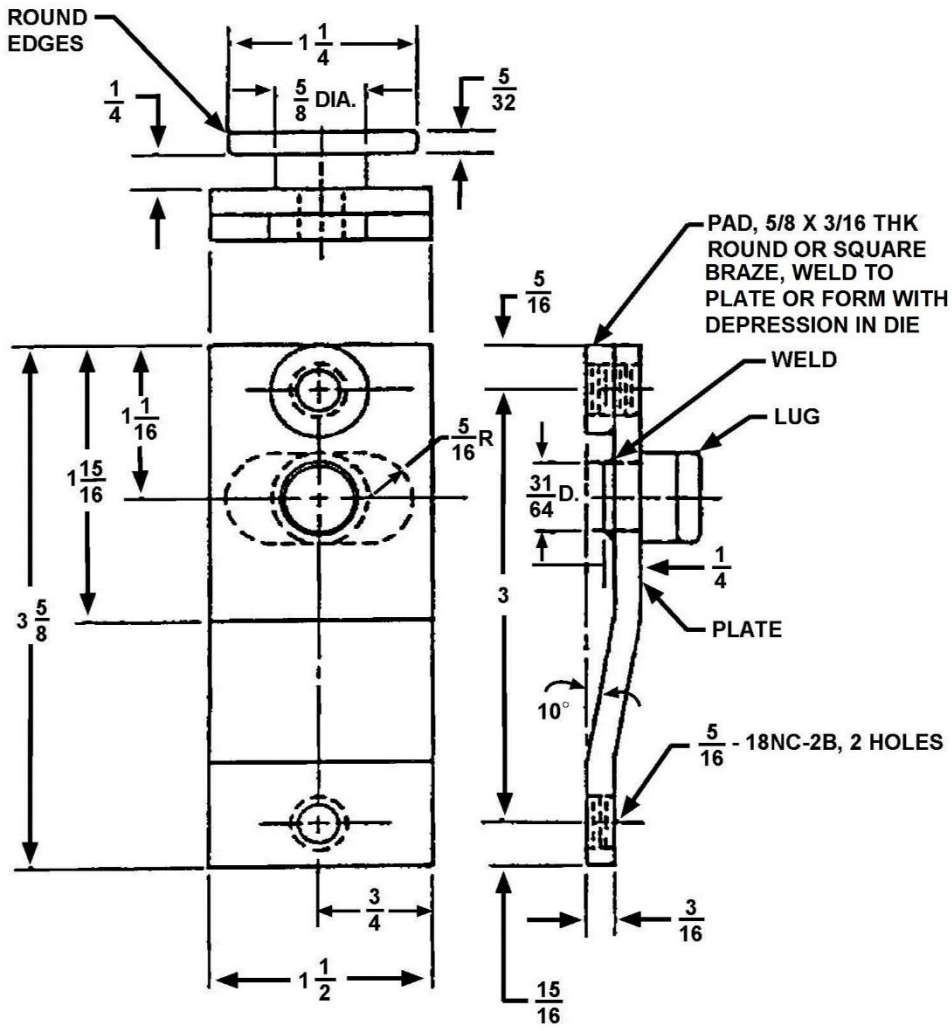


FIGURE 8. Universal bracket assembly.

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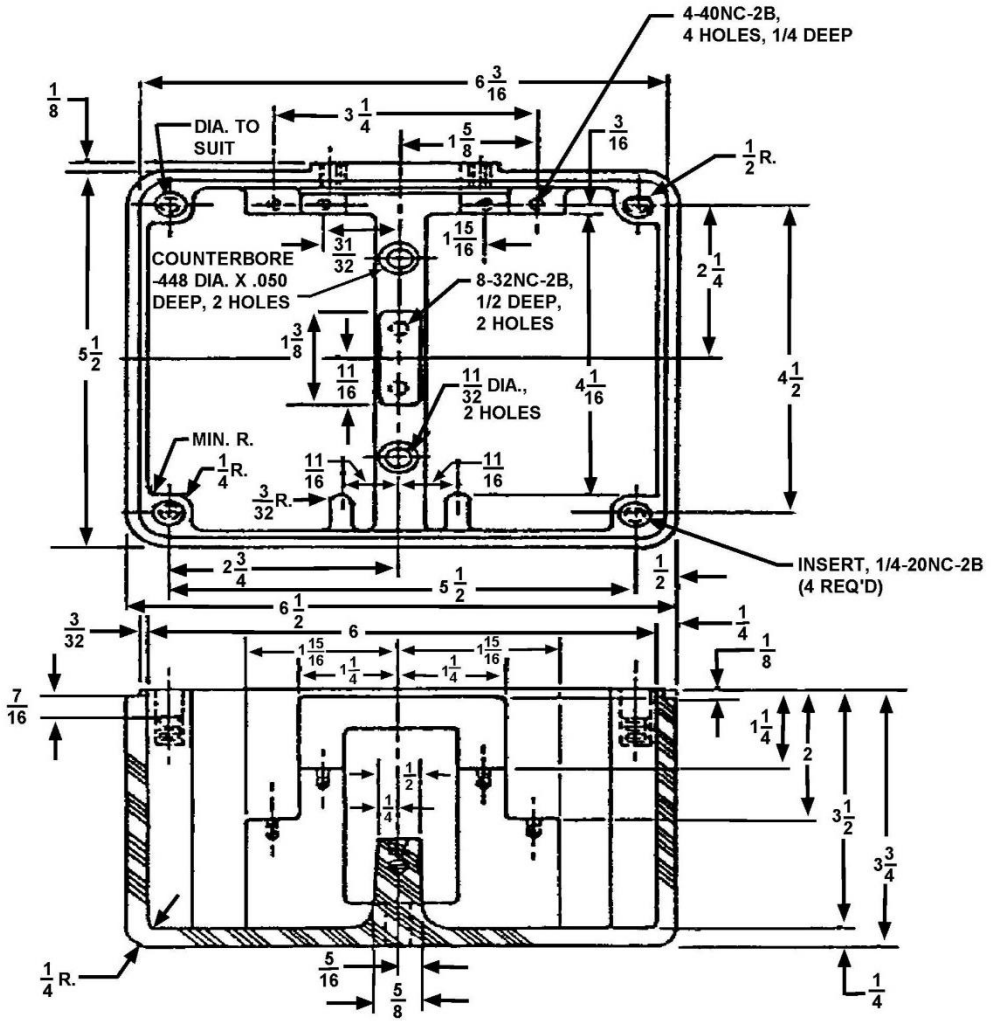


FIGURE 9. Universal box.

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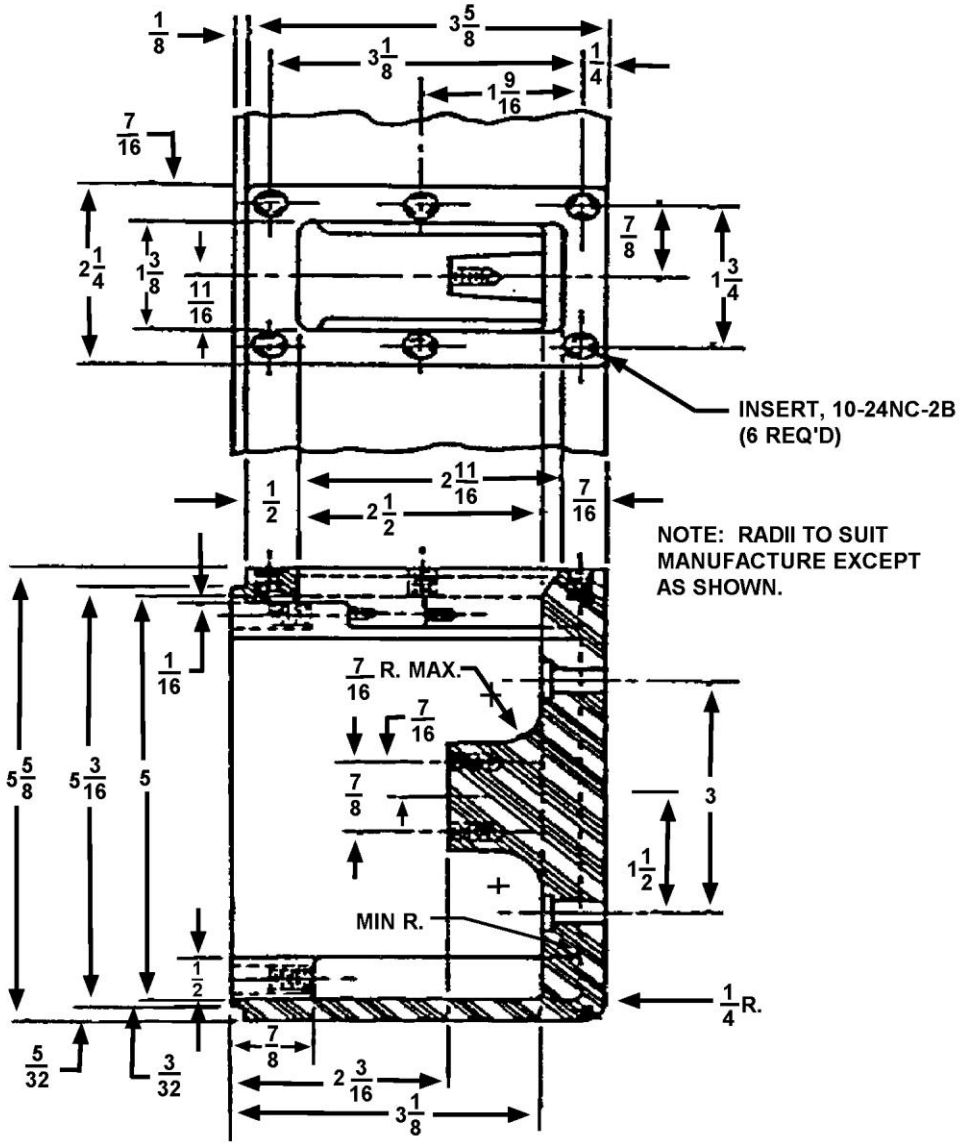


FIGURE 9. Universal box – Continued.

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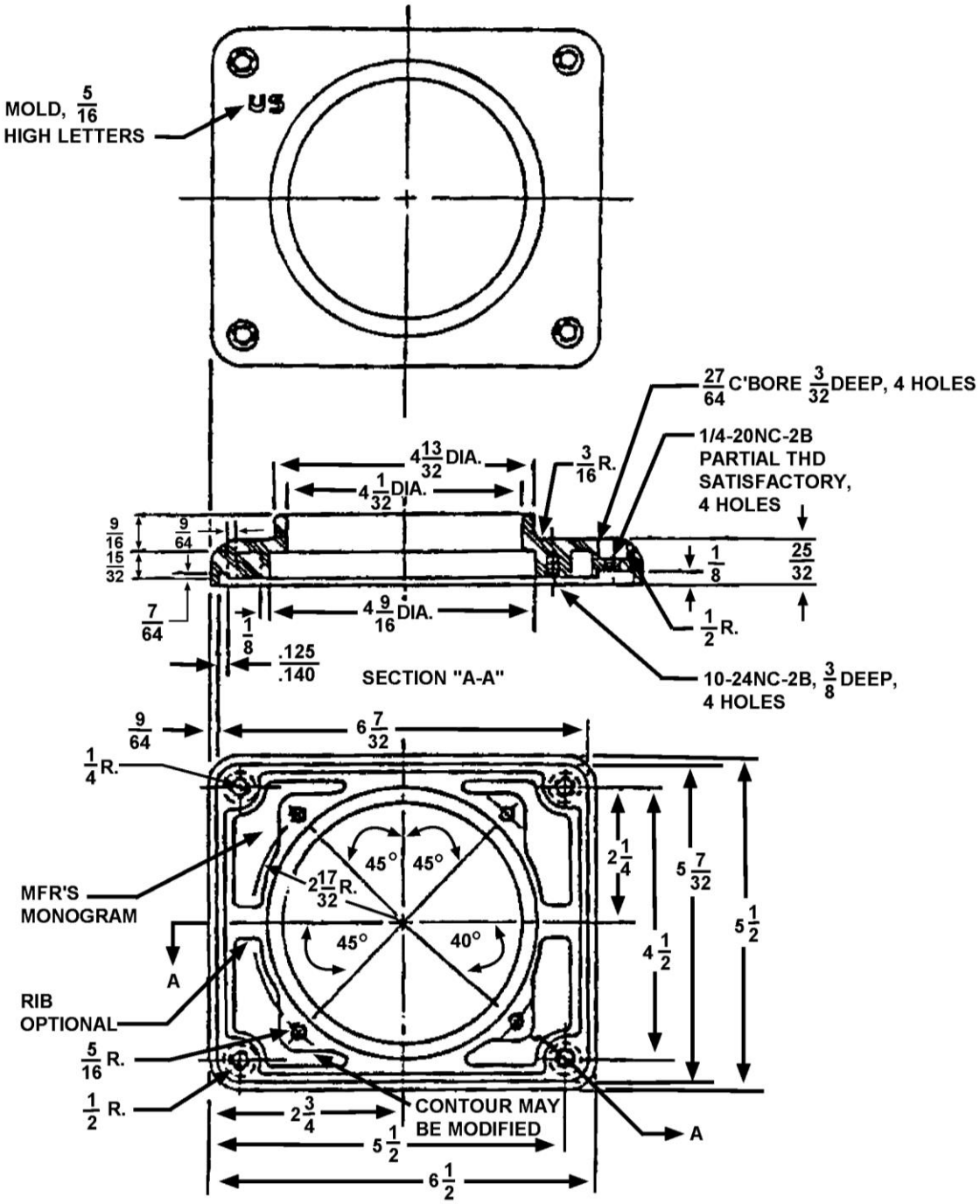


FIGURE 10. Universal cover.

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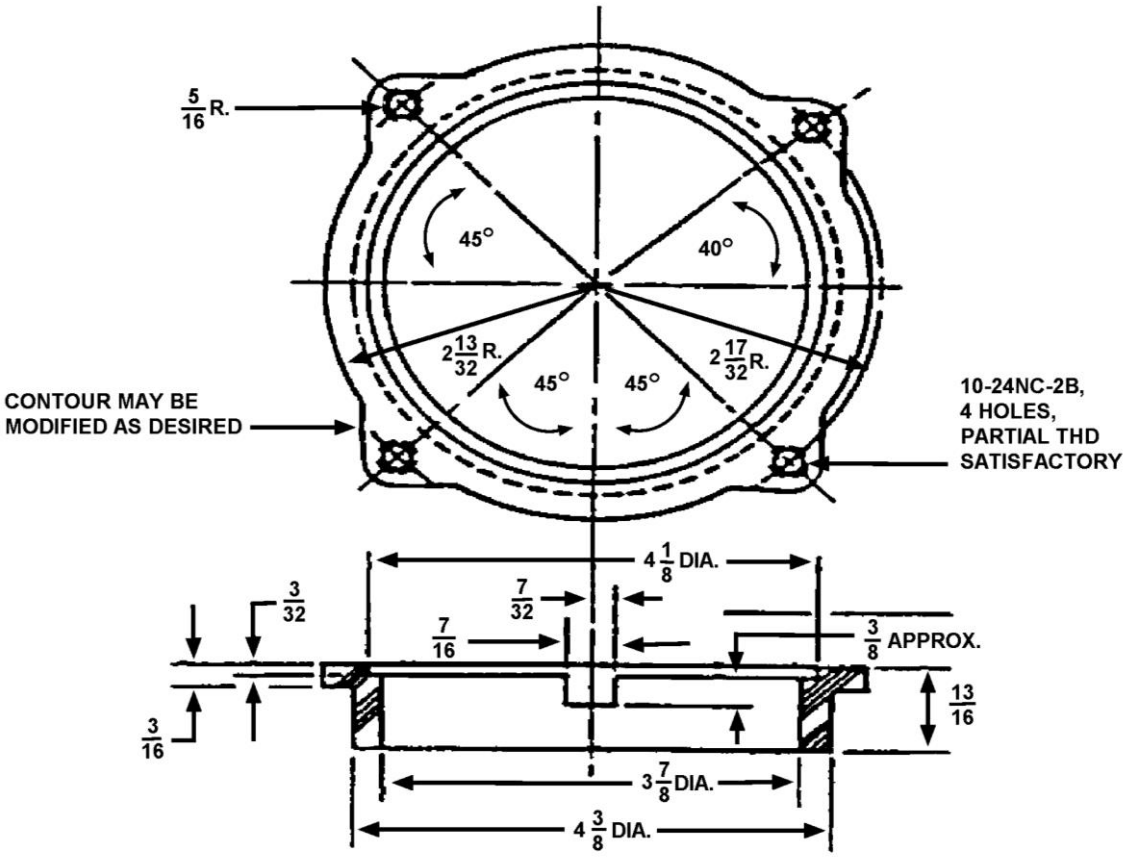


FIGURE 11. Universal lamp retainer.

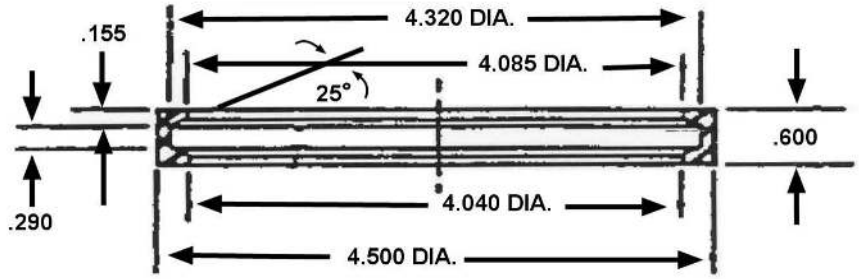


FIGURE 12. Gasket, LED assembly.

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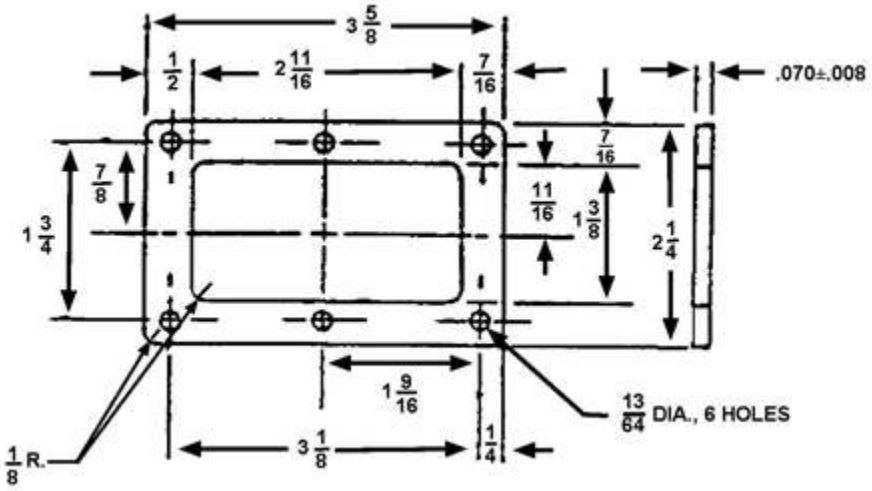


FIGURE 13. Universal gasket for handle and top plate assembly.

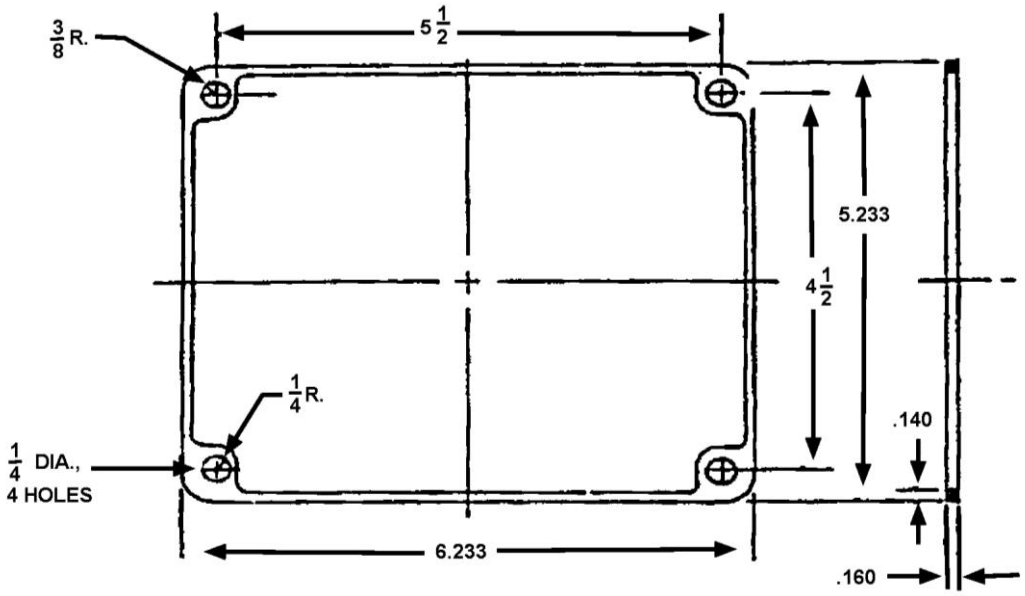


FIGURE 14. Universal gasket, cover.

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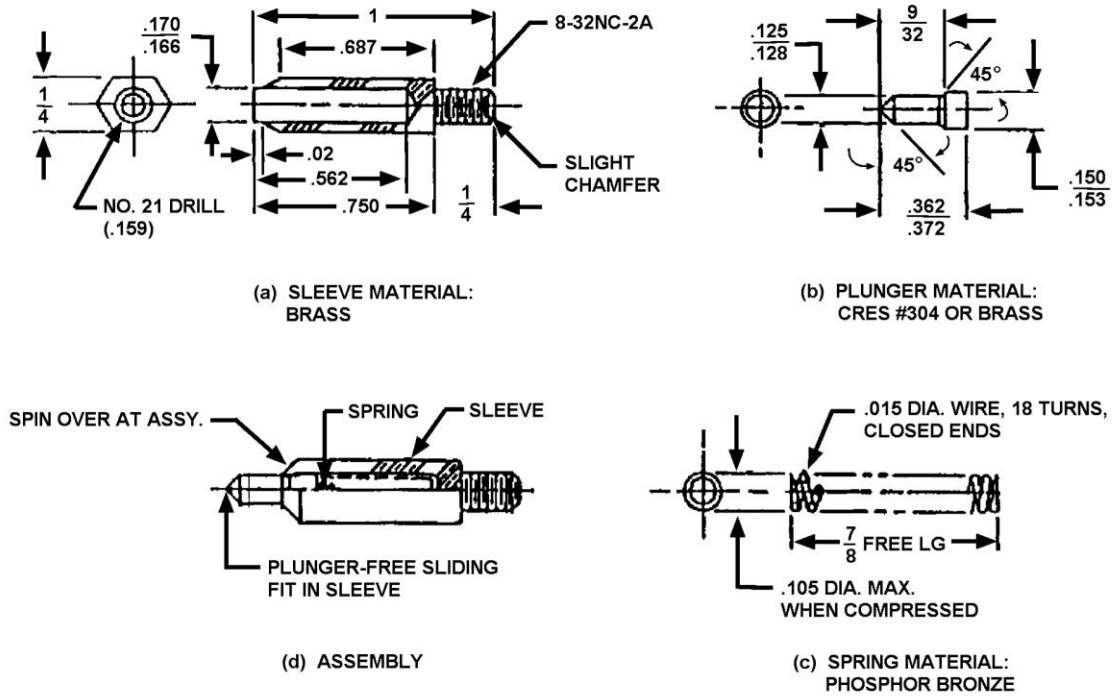


FIGURE 15. Universal lamp contact assembly.

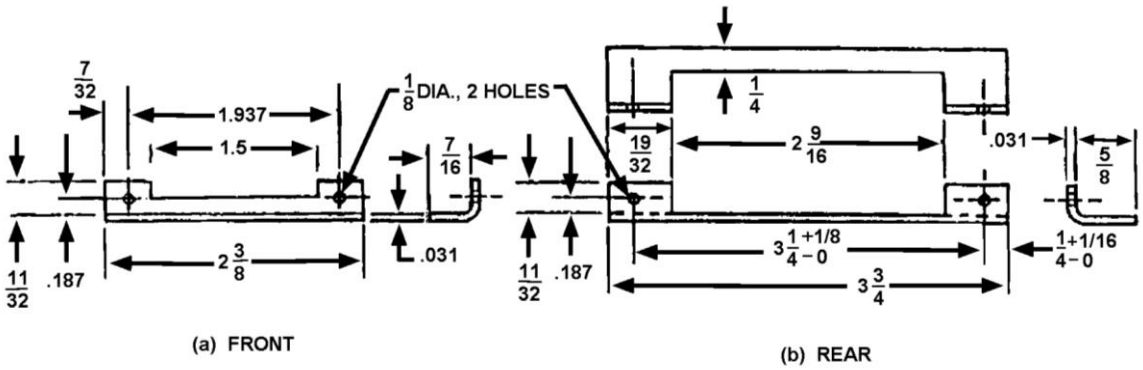


FIGURE 16. Universal battery contact connectors.

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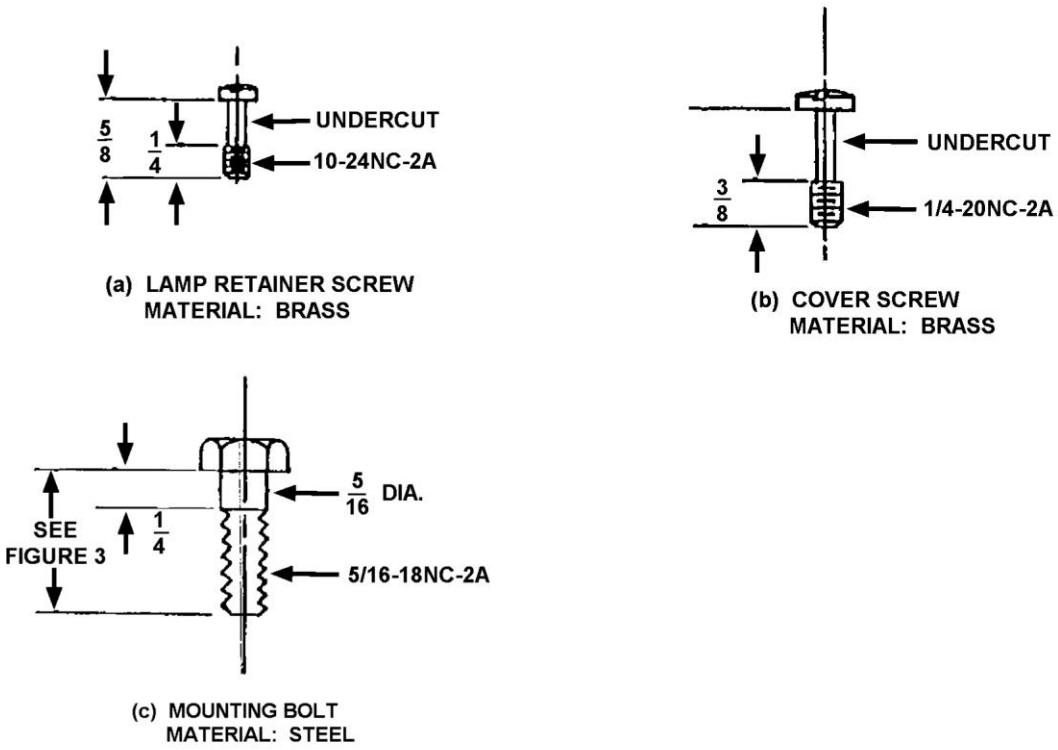


FIGURE 17. Universal screws and bolts.

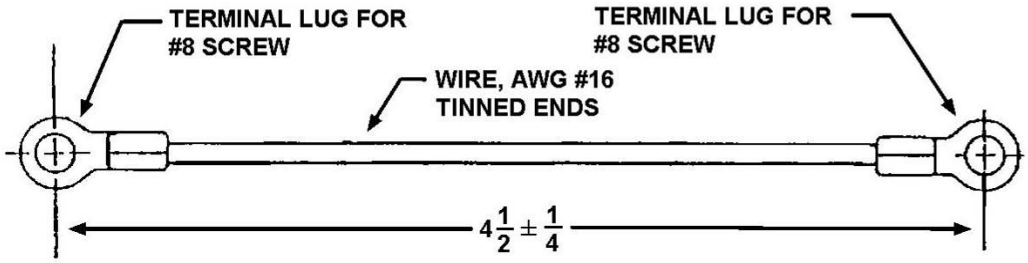


FIGURE 18. Universal wire assembly.

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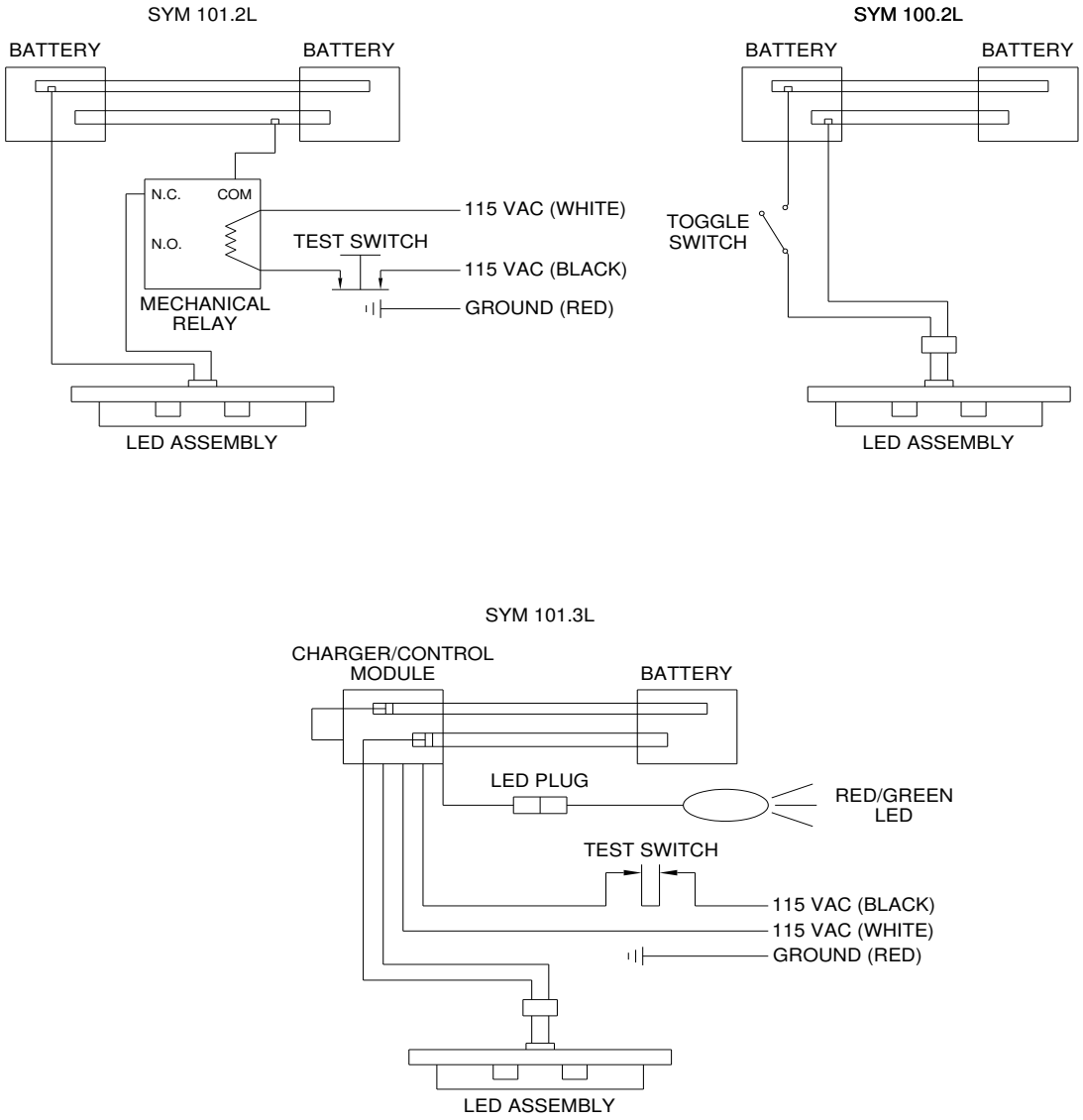
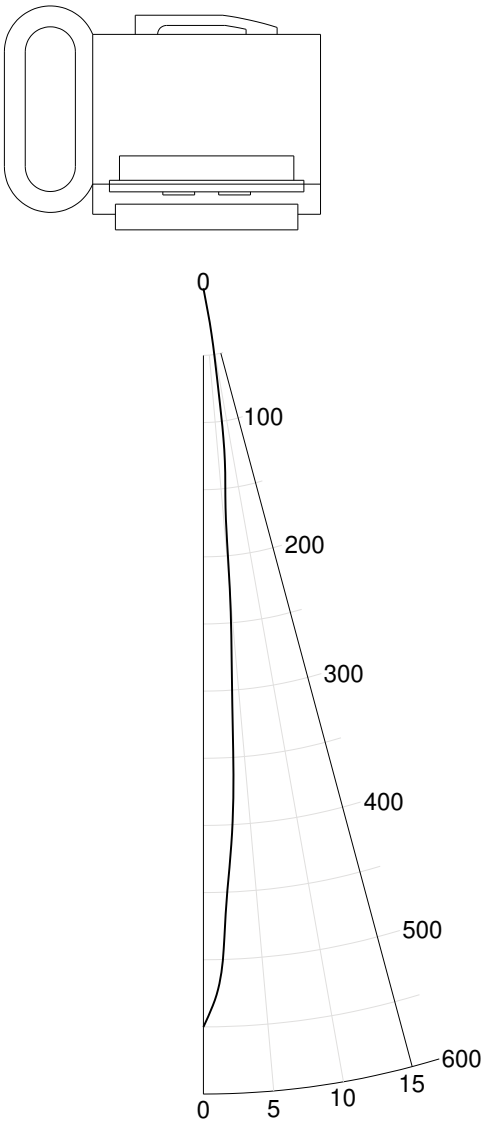


FIGURE 19. Wiring diagram.

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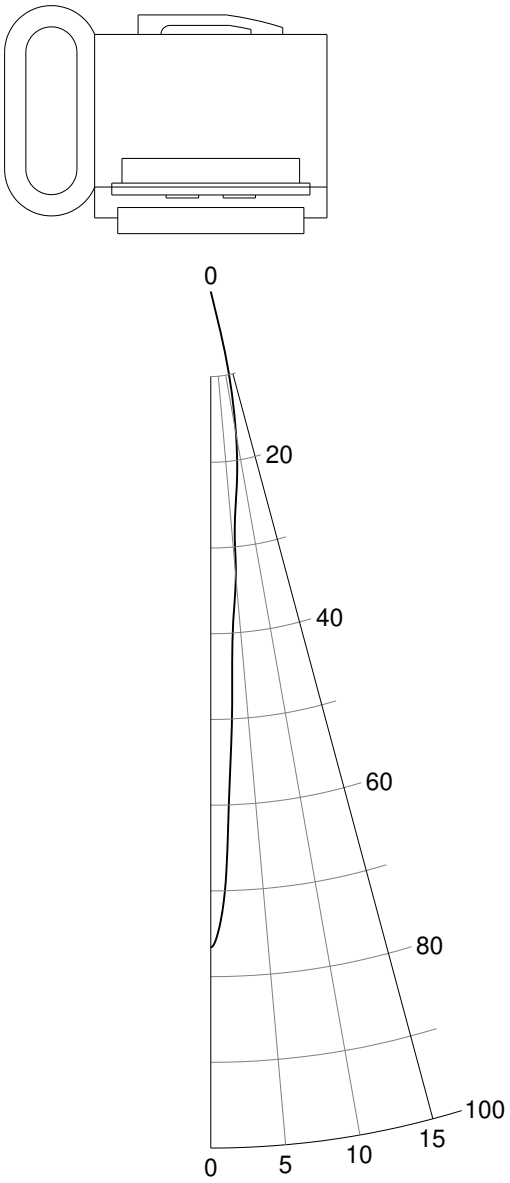


DEG	CANDELA
0	550
1	528
2	469
3	407
4	308
5	226
6	156
7	132
8	95
9	57

NOTE: Test distance is 10 feet.

FIGURE 20. White LED photometric test setup, resultant curve, and candela distribution.

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DEG	CANDELA
0	76
1	71
2	58
3	48
4	42
5	35
6	29
7	25
8	22
9	17

NOTE: Test distance is 10 feet.

FIGURE 21. Red LED photometric test setup, resultant curve, and candela distribution.

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REQUIREMENTS

Type: III
Class: 1

SYMBOL NUMBERS

100.2L: Lantern, portable, with handle and mounting bracket/LED model.
101.2L: Lantern, bulkhead, relay, 115 VAC/LED model.
101.3L: Lantern, bulkhead, rechargeable, 115 VAC/LED model.

DESIGN

Dimensions and configurations: See figure 1 through figure 21.
Material:
Box, cover, lamp retainer: Material for box and cover lamp retainer shall be molded plastic. The color of molded parts shall be yellow, conforming to color number 13591 of SAE AMS-STD-595, attained without resort to painting, enameling, or other post-molding applications. Plastic material shall be in accordance with ASTM D5948, type MAI-60.
Screws: All screws shall be brass.
Lockwashers: All lockwashers shall be phosphor-bronze, internal-lock.
Inserts: All inserts shall be brass.
Lantern assembly (complete) and typical mounting: See figures 1 and 2.
Body assembly: See figure 3. All parts shown on figure 3 shall be furnished assembled as indicated.
Bolts: Gaskets or bolts with self-contained gaskets shall be furnished to seal the mounting holes against the entrance of water where the body is assembled and mounted in the condition shown on figure 2.
Mounting bolts: Mounting bolts shall be as shown on figures 3 and 17.
O-ring: O-ring gaskets shall be furnished to seal the mounting holes against the entrance of water when the body is assembled and mounted in the condition shown on figure 2.
Lamp contact assembly: See figure 15. The plunger shall work freely in the sleeve without rubbing or coming adrift.
Battery contact connectors: See figure 16. Material shall be brass, half hard. Battery contact connectors shall be silver plated.
Box, cover, lamp retainer: See figures 9, 10, and 11, respectively.
Wire assembly: See figure 18.
Gasket, LED assembly: See figure 12. Material shall be neoprene rubber, 45 to 55 durometer A.
Gasket, cover: See figure 14. Material shall be neoprene rubber, 45 to 55 durometer A.

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LED assembly:	The LED cluster shall be balanced by electronic circuitry to provide evenly distributed light output in accordance with LED manufacturer specifications. The output color of the LED cluster shall be white or red, as specified. The LED cluster shall provide luminance equal to or greater than that included on figure 20 for white lanterns or figure 21 for red lanterns. Chromaticity shall be in accordance with 3.7.4 of MIL-DTL-16377. The LED cluster power terminals or wiring shall indicate polarity for proper installation. The LED assembly shall consist of a plexiglass or plastic lens in accordance with MIL-DTL-917 and MIL-DTL-24191 and an LED mounted on a backplate that will fit within the universal lamp retaining assembly (see figure 11). The assembly shall also provide a watertight seal to preserve the watertight integrity of the lantern.
Time of LED illumination (for symbol 101.3L):	Rechargeable battery shall maintain the LED cluster illuminated for not less than 32 hours from the loss of 115 VAC power before the low voltage cutoff of 5.25 to 5.5 V is reached.
Portable lantern handle assembly:	See figure 4 for symbol 100.2L. All parts shown on figure 4 shall be furnished assembled as indicated.
Switch:	Switch shall be military part number MS16656-2.
Switch boot:	Boot shall be in accordance with MIL-DTL-5423/2.
Gasket:	See figure 13. Material shall be neoprene rubber, 45 to 55 durometer A.
Bulkhead-mounted lantern top plate assembly:	See figure 5 for symbol 101.2L. See figure 6 for symbol 101.3L. All parts shown on figure 5 or figure 6 shall be furnished assembled as indicated.
Packing assembly:	Packing assembly shall be in accordance with ASTM F1836M.
Test switch:	Test switch shall be pushbutton, momentary off, meeting the below requirements.
Nominal voltage:	120 VAC.
Current rating at nominal voltage:	6 amperes minimum.
Electrical life:	25,000 cycles minimum.
Mechanical life:	100,000 cycles minimum.
Operating temperature:	32 to 185 °F (0 to 85 °C).
Thread:	15/32 UNS-2A.
Boot:	Boot shall be MIL-DTL-5423/7, style 2, size 3.
Gasket:	See figure 13. Material shall be neoprene rubber, 45 to 55 durometer A.
Cable:	Cable shall be 10 feet of type LSTHOF-3 in accordance with MIL-DTL-24643/3. The black and white conductors shall be used as the 115 VAC line inputs. The red conductor shall be used as the ground conductor. See figure 19 (wiring diagram).
Bracket assembly:	See figure 8. Material shall be steel, zinc plated.
Wire assembly:	See figure 18.
Ground connection:	The metal neck of the test switch and the relay metallic frame (if used) shall be electrically connected to the red lead of the supply cable for grounding to ship's structure.
Wiring:	Interconnecting and power leads shall be connected to relay and switch terminals using solder or crimp-style connectors. Wire nuts shall not be used.

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Insulating sleeve:	Shrink-on insulating sleeves shall be installed over all electrical connections carrying 115 or higher voltage. No metallic parts of such connections shall be exposed. Insulating sleeves shall not be loose or allowed to move. Insulating sleeves shall be placed over all unused contacts of the relay in the symbol 101.2L lantern. Insulating sleeves are not required for ground connections or electrical connections of the battery circuit.
Relay:	Relays shall conform to the requirements of table I and to the tests of table II. Any metallic case material shall be grounded to the red conductor. All unused contacts shall be insulated with an insulating sleeve.
Rechargeable retrofit kit:	The rechargeable retrofit kit shall convert a bulkhead-mounted symbol 101.2L (non-rechargeable) lantern into a symbol 101.3L (rechargeable) lantern. The rechargeable retrofit kit shall consist of a charger/control module, a rechargeable top plate (see figure 6) with all required parts (except cable) and the appropriate wiring and connectors. The rechargeable battery specified herein shall be acquired separate from the rechargeable retrofit kit. The configuration of the charger/control module and the rechargeable battery shall be such that they cannot be placed in the incorrect position. Appropriate wiring for the charger/control module and the LED indicator assembly shall be supplied as part of the rechargeable retrofit kit.
Charger/control module (for symbol 101.3L only):	
Power input:	115 VAC.
High battery voltage cutoff:	6.9 V nominal.
Low battery voltage cutoff:	5.25 to 5.5 V nominal.
Maximum charging current:	500 mA.
Recharging activation:	The charger/control module shall automatically initiate charging once the battery has less than 90 percent of full charge and 115 VAC power is restored. Recharging shall terminate at the high battery voltage cutoff.
Lantern activation:	The charger/control module shall automatically turn the lantern on when there is a loss of 115 VAC power and remain on until the battery reaches the low voltage cutoff or power is restored.
Maximum recharge time:	16 hours.
Electromagnetic interference:	In accordance with RE101, RE102, CE101, and CE102 requirements of MIL-STD-461.
Terminals:	Shall connect to/make contact with the existing battery contact connectors (see figure 19).
Dimensions:	The charger/control module shall have identical dimensions of a type BA-200/U battery in accordance with ANSI C18.1M.
Red/green LED indicator (for symbol 101.3L):	The red/green LED indicator shall be mounted on the rechargeable top plate (see figure 6) such that it is readily visible from the front of the lantern. The LED indicator shall have the following five modes of operation: No illumination: No input power to the lantern. Solid green: Battery is fully charged. Blinking green: Battery is charging. Solid red: Charging circuit has failed. Blinking red: Battery failure.
Wiring/connections (for symbol 101.3L):	The connections between the power input leads and charger/control module and the connections between the LED indicator assembly and the charger/control module shall be made using crimp-style connectors.

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Batteries (for symbols 100.2L and 101.2L):	Lanterns shall accommodate two batteries in parallel, type BA-200/U in accordance with ANSI C18.1M. Batteries are not furnished.
Rechargeable battery (for symbol 101.3L):	Rechargeable lanterns shall accommodate one rechargeable battery as specified herein. Batteries are not furnished with the rechargeable retrofit kit.
Type:	Sealed, lead-acid, maintenance free.
Nominal voltage:	6.0 V.
Minimum capacity:	4.0 ampere-hours at 20-hour rate, load of 200 mA. 3.0 ampere-hours at 6-hour rate, load of 500 mA.
Terminals:	Two coil springs, for contact with the existing battery contact connectors (see figure 15).
End of discharge voltage:	5.5 V nominal.
Self-discharge per month at 68 °F (20 °C):	Not greater than 3 percent.
Operating temperature range:	0 to 122 °F (-17.78 to 50 °C).
Cycle life (at 500 mA, 100 percent depth of discharge):	Not fewer than 800 cycles before battery is down to 80 percent of rated capacity.
Capacity loss per 100 cycles:	Not greater than 2.5 percent.
Dimensions:	The rechargeable battery shall have the identical dimensions of a BA-200/U type battery of ANSI C18.1M.
Enclosure:	Watertight-complete lanterns.
Lantern operational ambient temperature range:	32 to 122 °F (0 to 50 °C)
Maximum weight:	
Symbol 100.2L:	5 pounds, 9 ounces maximum, without batteries.
Symbol 101.2L	6 pounds, 8 ounces maximum, without batteries.
Symbol 101.3L:	10 pounds, 8 ounces maximum with rechargeable battery, charger/control module, and handle on top plate, with the red/green battery status LED indicator installed.
Labels:	Labels shall be furnished in accordance with MIL-DTL-16377. The following labels and information shall be furnished.
Type I:	See MIL-DTL-16377.
Type III:	Wiring diagram of figure 19.
Type IV:	"USE FOLLOWING REPAIR PARTS: Boot for toggle switch: M5423/02 Boot for pushbutton switch: M5423/07-01 Toggle switch: MS16656-2 Pushbutton switch: M16377/53-015"

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EXAMINATION AND TESTS

Examination and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Body assembly:	See table II.
Handle assembly:	See table II.
Rechargeable retrofit kit:	See table II.
LED assembly:	See table II.
Bracket assembly:	Examination only.
Conformance:	
Comparison inspection:	Same as first article inspection.
Conformance inspection:	See MIL-DTL-16377.

GENERAL INFORMATION

Assignment of military part numbers and general information is shown in table III.

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TABLE II. Required first article testing for battle lanterns.[1/]

Test	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	2/, 3/
Insulation resistance	3.6.16	4.8.3	2/, 3/
Enclosure effectiveness	3.5.11	4.8.14.4	2/, 4/
High-impact shock	3.5.12	4.8.7	Grade A2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14.4	2/, 4/
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Mechanical shock	3.7.7.6	4.8.17.3	8 foot-pounds
Thermal shock (glassware/plastics)	3.7.7.7	4.8.17.4	---
Power interface	3.6.18	4.8.21	3/
Light output	3.7.1	4.8.6.1	Figure 20 (white) or figure 21 (red)
Chromaticity	3.7.4	4.8.6.5	---
Electromagnetic interference	3.6.14	4.8.16	5/
High ambient temperature operation	3.5.17	4.8.26	6/

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

3/ Test required only for symbols 101.2L and 101.3L and the rechargeable retrofit kit.

4/ Watertight.

5/ Charger/control module only. Only tests RE101, RE102, CE101, and CE102.

6/ 122 °F ambient temperature; applies to symbol 101.3L and rechargeable retrofit kit only.

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TABLE III. Part number and general information.

Military Part Number M16377/53	Item Name	Remarks
100.2L	Complete lantern, symbol 100.2L	1/
101.2L	Complete lantern, symbol 101.2L	2/
101.3L	Complete lantern, symbol 101.3L	3/
001	Body assembly	---
002	Handle assembly	---
003	Bracket assembly	---
004	115 VAC relay assembly	---
005	No longer used	---
006	No longer used	---
007	Cover, gasket, figure 14	---
008	Rechargeable retrofit kit, bulkhead	Includes rechargeable top plate assembly
009	Rechargeable battery	Purchased separately
010	LED assembly, white	---
011	No longer used	---
012	No longer used	---
013	115 VAC rechargeable; top plate assembly	Included in rechargeable retrofit kit, bulkhead
014	LED assembly, red	---
015	Switch, pushbutton	---

NOTES:

1/ Symbol 100.2L consists of part numbers M16377/53-001, M16377/53-002, M16377/53-003, and M16377/53-010 or -014

2/ Symbol 101.2L consists of part numbers M16377/53-001, M16377/53-004, and M16377/53-010 or -014.

3/ Symbol 101.3L consists of part numbers M16377/53-001, M16377/53-008, M16377/53-009, M16377/53-013, and M16377/53-010 or -014.

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – AV
Navy – SH

Preparing activity:
Navy – SH
(Project 6230-2021-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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